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Predictive Modelling of Country-Level Development Using Explainable Machine Learning

Submitted in partial fulfilment for the award of the degree of

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by

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CERTIFICATE

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I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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ABSTRACT

This project examines how machine learning can model and predict important socio-economic development indicators, such as the Human Development Index (HDI), GDP per capita, and Happiness Score. Using structured country-level data from the World Bank and global well-being datasets, the project looks at how different factors, like access to electricity, education and health spending, social support, and views on corruption, affect a country's development status.

We used a multi-output Random Forest regression model with the MultiOutputRegressor wrapper to predict multiple targets at the same time. We trained and evaluated the model using R^2, Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE), which provided a solid baseline for performance. To ensure clarity, we used SHAP (SHapley Additive explanations) to assess feature importance and support clear, data-driven insights.

Drawing from recent studies on explainable AI and policy simulation, this work also includes "what-if" scenario modeling. This allows us to explore how specific changes, such as boosting health spending or lowering corruption, can affect development outcomes. The results show the potential of interpretable machine learning models to help guide policy-making and improve decision-making for sustainable development.

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Place: Bangalore

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MANET	Mobile Ad hoc Network
WAP	Wireless Access Point

Chapter 1: Introduction

1.1 INTRODUCTION

Every country wants to grow not just economically, but in ways that improve the lives of all its people. But development is a complicated puzzle, influenced by dozens of social and economic factors like education, health, infrastructure, and income levels. Understanding how these pieces fit together, and more importantly, how changing one affects the others, is a challenge faced by policymakers and researchers around the world.

That’s where machine learning can help. With the power to uncover patterns and simulate possible futures, it gives us a way to ask important "what-if" questions like what might happen if a country invests more in education, or focuses on reducing inequality.

1.2 PROBLEM STATEMENT

Even though we have access to a wealth of development data, it's often challenging to know what to do with it and of which factors matter most? Where should a country focus its energy and resources? And how do different interventions impact not just one area of development, but many? Traditional analysis can only go so far, especially when trying to understand such interconnected systems.

1.3 OBJECTIVES

The main goal of this project is to build a machine learning model that can learn the relationship between various socio-economic inputs and multiple development outcomes. We would want to make a difference in:

- Explore how different development indicators are related to each other.
- Simulate "what-if" scenarios to guide policy or investment decisions.
- Compare the effectiveness of different machine learning models (like multi-output neural networks and random forests in solving this problem.
- Ultimately, highlight which areas a country should focus on to achieve well-rounded development.

1.4 SCOPE OF THE PROJECT

In this project, we focus on building and evaluating two models: a multi-output neural network and a random forest regression model to predict multiple development indicators based on socio-economic data. We then compare their performance to understand which model works better for this type of problem.

The idea is not just to get accurate predictions, but to gain insight into how countries can prioritise their development efforts. This could be useful for policymakers, researchers, and citizens who care about making data-driven decisions for a better future.

Chapter 2: Background

Understanding a country's development goes beyond just looking at economic performance. GDP is a common measure, but it doesn't capture important factors like income inequality and citizen well-being. To get a fuller view, we chose three key indicators: Human Development Index (HDI), GDP per capita, and the Happiness Score.

2.1 Why These Indicators?

HDI: This composite index measures life expectancy, education, and income. It reflects overall human well-being.

GDP per capita: This shows the average economic output per person, but it doesn't indicate how income is distributed.

Happiness Score: This is based on surveys that assess social support, freedom, corruption perception, and more. It reflects personal well-being.

Together, these indicators cover economic, social, and emotional aspects of development.

2.2 Why Not Traditional Analysis?

Traditional models often concentrate on one indicator and make simple correlations. They overlook:

Complex, non-linear relationships between variables.

Interactions between inputs, such as education and healthcare.

Clear feature importance or policy recommendations.

2.3 What Makes Our Approach Unique

We use a Multi-Task Neural Network (MLP) that:

Predicts all three indicators together.

Shares learning across outputs for improved performance.

Utilizes SHAP values to explain which features, like education and healthcare, influenced each output.

Supports "what-if" scenario simulations for planning policies.

2.4 Insights from Literature Survey

Research shows that education, health, and governance strongly affect development. Machine learning models like Random Forests and MLPs can effectively predict indicators such as HDI and GDP, while explainability tools like SHAP help interpret the results. These findings back our model design and objectives.

2.4 Evaluation Metrics Used

MSE: This penalizes large prediction errors.

R^2 Score: This indicates how well the model explains the data.

MAPE: This measures percentage-based prediction error.

SHAP: This explains which features had the most impact on each prediction.

Chapter 3: Methodology

Neural Networks

3.1 Dataset Selection

We used the World Bank's World Development Indicators (WDI) dataset, which provides time-series, country-level data in key areas like health, education, governance, and infrastructure.

3.2 Feature Selection

Our target variables included:
- HDI
- GDP per capita
- Happiness Score

Initial input features (based on existing research):

For HDI/GDP: Access to electricity, education and health spending, and R&D spending.
For Happiness: Social support, freedom, generosity, and perception of corruption.

Final input features (after trials and SHAP analysis):
All of the above, plus HDI-specific features such as life expectancy, years of schooling, and GNI per capita.

3.3 Data Preprocessing

The dataset needed cleaning and restructuring.

The original wide-format data, with years as columns, was reshaped into a long format using pandas.melt()

We addressed missing data in the following ways:

- Used forward fill for continuous variables
- Used backward fill for gaps
- Deleted rows when data was critically insufficient

We also normalized features to stabilize training across different scales, like percentage values and GDP in billions.

3.4 Machine Learning Models

Neural Network (Multi-Task MLP):

We built a Multi-Output Feedforward Neural Network using Keras. This model:

Takes socio-economic indicators as inputs.

Outputs predictions for HDI, GDP per capita, and Happiness Score.

Uses shared hidden layers to learn interactions between outputs.

Why does it fit our project:
Neural networks can capture non-linear relationships between complex socio-economic inputs and multiple outputs. Multi-task learning also shares useful patterns across outputs, which improves generalisation.

3.5 Model Evaluation

We evaluated the models using:

MSE (Mean Squared Error), which measures the squared prediction error.

R^2 Score, which shows how well the model explains output variance.

MAPE (Mean Absolute Percentage Error), which quantifies relative error in percentage terms.

3.6 Explainability with SHAP

To interpret the model's decisions, we used SHAP (SHapley Additive explanations). SHAP assigns a contribution value to each input feature for each prediction. It helps answer questions like:

"Which features most influenced GDP in Country X in 2020?"

"How does increased education spending affect HDI?"

This improves transparency and trust. It also supports data-driven policy decisions.

3.7 What-If Scenario Simulation

After training the model, we ran what-if simulations like these:

Increasing health spending by 1%

Improving electricity access by 10%

Reducing perceived corruption

These scenarios demonstrate how such changes could affect HDI, GDP, and Happiness. They help pinpoint high-impact investment areas for governments and decision-makers.

3.8 Tools and Libraries

Python (pandas, numpy) – data processing
Scikit-learn – baseline models
Matplotlib / Seaborn / Plotly – visualisations
SHAP – explainable AI analysis

Random Forest

3.1 Dataset Selection

We continued using the World Bank's World Development Indicators (WDI) dataset, which offers rich, country-level, time-series data covering key developmental areas like health, education, governance, infrastructure, and economics.

3.2 Feature Selection

Our goal remained to predict:

- HDI (Human Development Index)
- GDP per capita
- Happiness Score

Initial Input Features:

For HDI/GDP: Access to electricity, education and health expenditure, R&D spending.
For Happiness: Social support, freedom, generosity, and corruption perception.

Final Input Features :
We retained all initial features and added HDI-related indicators like life expectancy, GNI per capita, and average years of schooling.

3.3 Data Preprocessing

The WDI dataset was initially in wide format (years as columns), which we reshaped into a long format using pandas.melt()

Steps we took to clean the data:

- Missing values:

  Forward fill for trends
  Backward fill for isolated gaps
  Row deletion when too sparse

- Normalization:
  Applied scaling on variables (like GDP, percentages) to avoid model bias due to magnitude differences.

3.4 Machine Learning Model: Random Forest Regressor

We trained a Random Forest Regressor using scikit-learn to predict all three target variables. This model offers a tree-based, ensemble learning approach and is known for:

Handling non-linear relationships well
Being less sensitive to outliers
Working robustly even with some missing data

We wrapped the RandomForestRegressor in a MultiOutputRegressor to support our multi-target setup.

Why Random Forest?

Compared to neural networks, Random Forests:

Require less hyperparameter tuning
Are faster to train on structured/tabular data
Offer built-in feature importance metrics

They're a great baseline and complement the neural model for comparison.

3.5 Model Evaluation

We evaluated the model on a holdout test set using:

MSE (Mean Squared Error) – Penalises large errors
R^2 Score – Tells us how much variance our model explains
MAPE (Mean Absolute Percentage Error) – Percentage error, useful across different value scales

These metrics were computed separately for each of the three output variables.

3.6 Explainability with SHAP

We used SHAP (SHapley Additive explanations) to demystify our Random Forest's predictions.

With SHAP, we could answer:

"What factors most affected GDP in 2019 for India?"
"How much did education spending contribute to HDI?"

This gave us transparent insights into the model's behaviour which is crucial for policy-related applications.

3.7 What-If Simulations

Using the trained Random Forest, we simulated several hypothetical interventions:

Boosting health spending by 1%
Increasing electricity access by 10%
Improving perception of freedom or reducing corruption

These simulations gave us grounded, quantitative answers to: "If we change X, how might HDI or Happiness shift?"

This makes the model useful for evidence-based development policy planning.

3.8 Tools and Libraries

Python – Core language
pandas, numpy – Data wrangling
scikit-learn – Random Forest, evaluation, multi-output handling
matplotlib / seaborn – Visualisations
SHAP – Model interpretability

Chapter 4: Implementation

Neural Networks

4.1 Datasets

We found the necessary dataset from Kaggle, the World Development Indicators (WDI) dataset from the World Bank, which provides standardised time-series data across over 200 countries.
The input features include:

- Access to electricity
- Education spending
- Health spending
- R&D spending
- Life expectancy
-

The target variables are :

- GDP per capita
- HDI
- MPI
- Happiness Score

4.2 Implementation

To tackle the multi-output regression problem, we used a feedforward neural network built with TensorFlow and Keras. Neural networks are well-suited for capturing complex, nonlinear relationships in data — especially when predicting multiple continuous outputs simultaneously.

We designed a simple yet effective neural network architecture with several dense (fully connected) layers. The final layer had four output nodes corresponding to our target variables: GDP per capita, Human Development Index (HDI), Multi-dimensional Poverty Index (MPI), and Happiness Score.

The key steps in our workflow were:

- Splitting the data into training and testing sets using an 80:20 ratio to evaluate generalisation
- Standardising the input features to ensure all variables had similar ranges, which helps the neural network converge faster and more effectively
- Training the model using the training data, optimising with the Adam optimiser and mean squared error (MSE) as the loss function
- Predicting all four development indicators simultaneously on the test set, leveraging the model's multi-output capabilities

4.3 Experimental setup

We have evaluated the neural network model using standard regression metrics:

- R^2 Score to measure how well the model explains variance
- MAE for average error size
- MAPE to understand errors in percentage terms across different indicators

The model performed well, capturing complex relationships between features and outputs. It also provides a strong baseline for exploring explainability tools like SHAP and running counterfactual “what-if” development scenarios.

Random Forest

4.1 Datasets

We used data from the World Bank's WDI and related sources for each country. The input features included:

- Access to electricity
- Education, health, and R&D spending
- Life expectancy
- Social support, freedom, generosity
- GNI per capita

The target variables were:

- GDP per capita
- Human Development Index (HDI)
- Multi-dimensional Poverty Index (MPI)
- Happiness Score

4.2 Implementation

To manage the multi-output regression task, we used Random Forest Regressors wrapped inside a MultiOutputRegressor from Scikit-learn. This approach allowed us to train a separate Random Forest model for each output variable while treating them as one unified task.

Key steps included:

- Splitting the dataset into training and testing subsets, typically in an 80:20 ratio
- Training the model using the training data
- Predicting all four output variables at the same time on the test data

We chose Random Forest because it performs well with tabular data and can model non-linear relationships without needing extensive preprocessing.

4.3 Experimental Setup

We have evaluated model performance using these metrics:

R^2 Score: This shows how well the model explains variance in the data.

Mean Absolute Error (MAE): This measures the average size of the errors.

Mean Absolute Percentage Error (MAPE): This gives a percentage-based error, which helps with understanding results across different indicators.

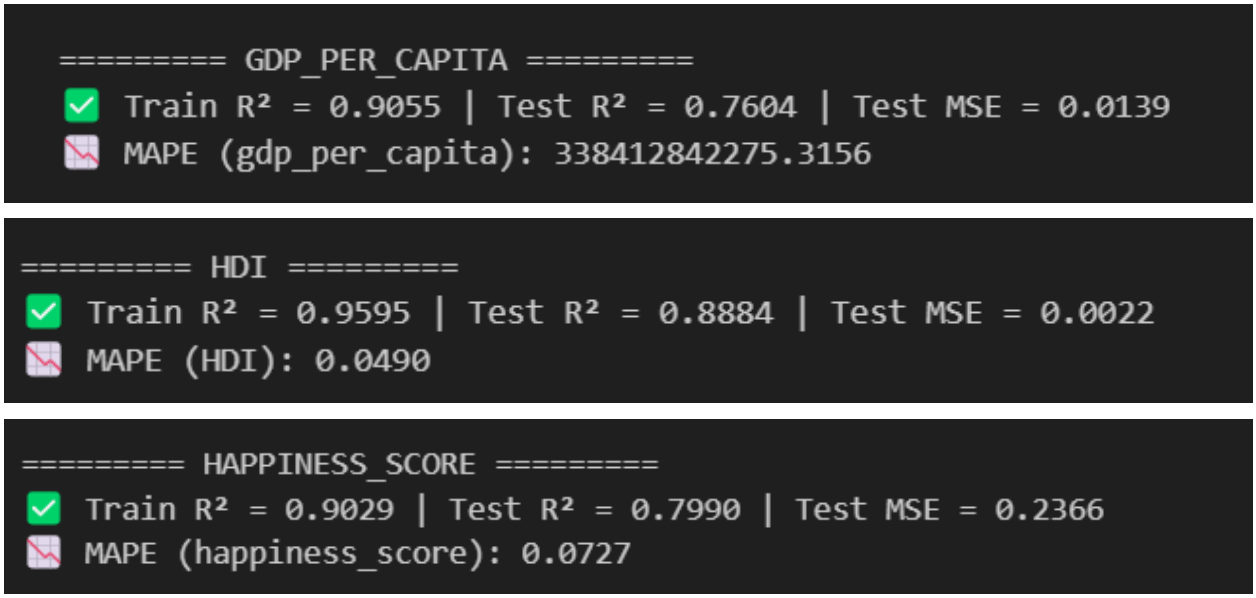
This baseline Random Forest model delivered strong predictive performance and acted as a reference for

comparison with more complex methods. It also set the stage for using Explainable AI techniques like SHAP to interpret results and simulate “what-if” development scenarios.

Chapter 5:Result

The results of this project primarily focus on the model performance of both models (Neural Network and Random Forest). While the initial expectation was that one model would largely outperform another, the final results were surprisingly similar.
The Neural Network model is more complex internally, which causes difficulty in identifying exactly what can be done to improve the model's performance. It is more sensitive to hyperparameters and overfitting.
The Random Forest model was relatively stable, exhibited less overfitting, and was also easier to understand for users unfamiliar with Machine learning concepts.

The output of the Neural network is as follows:



0.9 R^2 score is considered quite high. The decline in the R^2 score in the train and test data highlights the tendency of this model to overfit the data. During initial trails, this difference was a lot higher.

The Results of the Random Forest model are as follows:

```
--- gdp_per_capita ---
R² Score: 0.8237144365622322
MSE: 0.037712630055311 79
MAPE: 17795737480898.863

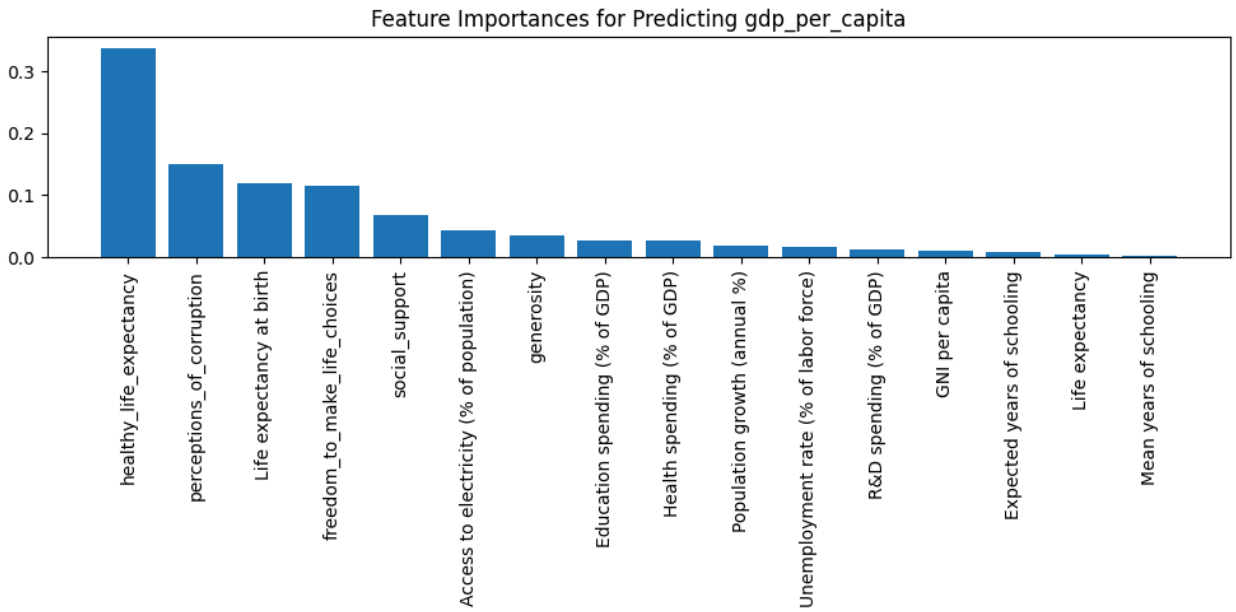
--- HDI ---
R² Score: 0.7954204818208607
MSE: 0.005062020638495345
MAPE: 0.07446442388433015

--- happiness_score ---
R² Score: 0.8614703566389608
MSE: 0.17871671327730862
MAPE: 0.06307963622829323
```

Lastly, the following feature importance was achieved :

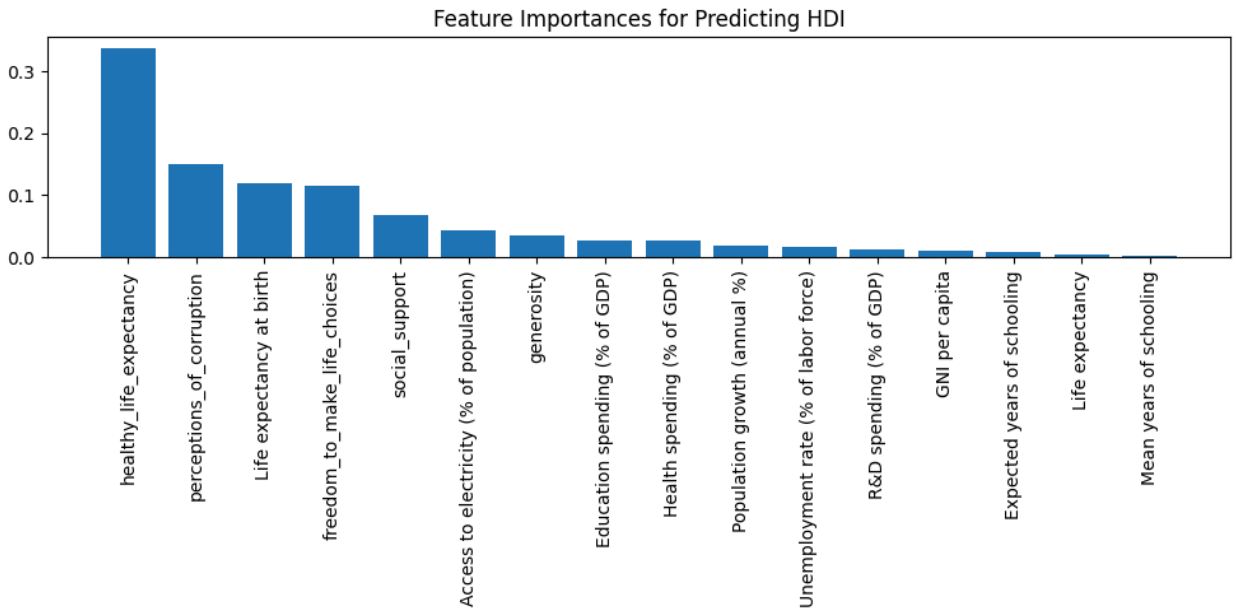
For GDP:

```
🔍 Feature Importance:
Life expectancy at birth                    0.625250
Access to electricity (% of population)     0.277508
healthy_life_expectancy                     0.264011
generosity                                  0.093253
Unemployment rate (% of labor force)        0.081421
Mean years of schooling                     0.069299
Expected years of schooling                 0.067428
GNI per capita                              0.060368
freedom_to_make_life_choices                0.055099
social_support                              0.054576
Life expectancy                             0.041737
R&D spending (% of GDP)                     0.041582
Health spending (% of GDP)                  0.040717
Population growth (annual %)                0.037438
perceptions_of_corruption                   0.030466
Education spending (% of GDP)               0.017893
```



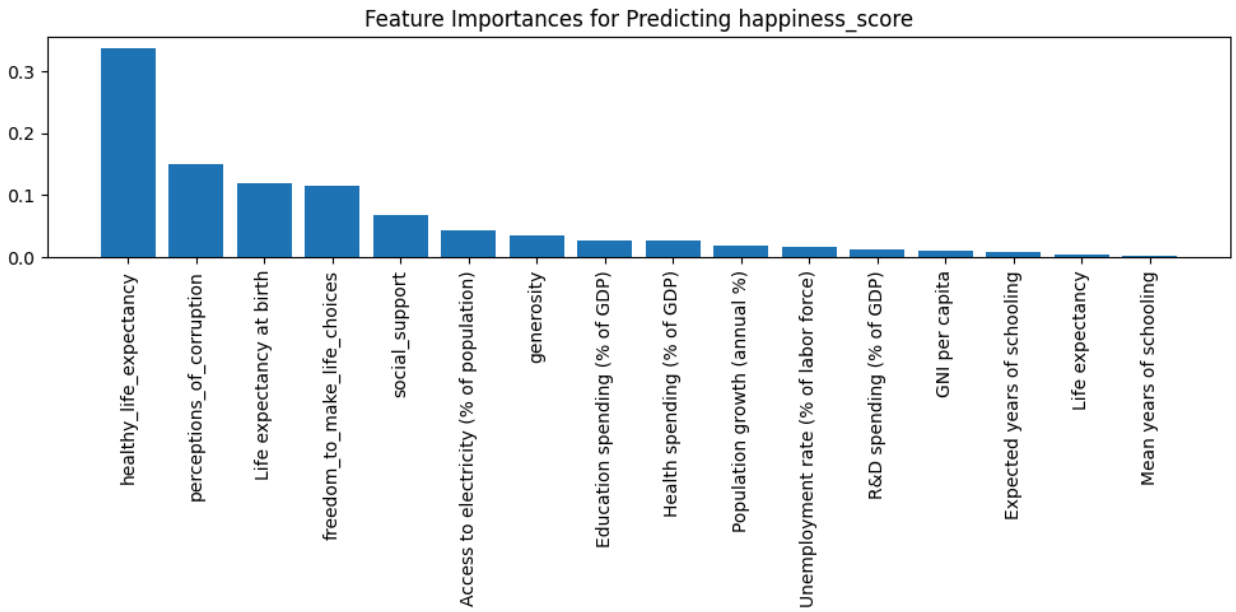
For HDI:

```
🔍 Feature Importance:
Life expectancy                          0.267948
Mean years of schooling                  0.215233
Expected years of schooling              0.170402
Health spending (% of GDP)               0.155852
healthy_life_expectancy                  0.146262
GNI per capita                           0.138000
Access to electricity (% of population)  0.098936
Life expectancy at birth                 0.090105
social_support                           0.082267
R&D spending (% of GDP)                  0.061420
perceptions_of_corruption                0.039385
freedom_to_make_life_choices             0.037514
Population growth (annual %)             0.027181
generosity                               0.019624
Education spending (% of GDP)            0.018231
Unemployment rate (% of labor force)     0.009042
dtype: float64
```



Happiness score:

```
🔍 Feature Importance:
Access to electricity (% of population)    0.563333
Life expectancy at birth                   0.295362
Life expectancy                            0.193609
GNI per capita                             0.093832
Health spending (% of GDP)                 0.089545
Population growth (annual %)               0.088393
R&D spending (% of GDP)                    0.083957
Unemployment rate (% of labor force)       0.072730
freedom_to_make_life_choices               0.069091
Mean years of schooling                    0.060647
social_support                             0.056858
healthy_life_expectancy                    0.043801
perceptions_of_corruption                  0.039338
Expected years of schooling                0.030352
generosity                                 0.023240
Education spending (% of GDP)              0.014007
dtype: float64
```


Feature Importances for Predicting happiness_score
0.3
0.2
0.1
0.0
healthy_life_expectancy
perceptions_of_corruption
Life expectancy at birth
freedom_to_make_life_choices
social_support
Access to electricity (% of population)
generosity
Education spending (% of GDP)
Health spending (% of GDP)
Population growth (annual %)
Unemployment rate (% of labor force)
R&D spending (% of GDP)
GNI per capita
Expected years of schooling
Life expectancy
Mean years of schooling

Chapter 6:Inference

In our comparative analysis, the Random Forest model stood out as a more stable and slightly more interpretable option, particularly when it came to predicting GDP and Happiness. This stability makes it a strong candidate for applications where understanding the model's reasoning is just as important as the prediction itself. Random Forests allow us to see which features influence the results, which is incredibly useful in socio-economic contexts where transparency and insight are essential. In contrast, the Neural Network model performed best when predicting Human Development Index (HDI). This suggests that HDI may be influenced by deeper, more complex nonlinear relationships—patterns that neural networks, with their layered architecture, are particularly good at capturing. While neural networks can sometimes feel like a black box, in this case, they provided a small but meaningful increase in accuracy where it mattered.

Interestingly, despite the differences in model behaviour, both approaches pointed to a common and critical insight: life expectancy consistently appeared as one of the most important predictors across all three indicators i.e, HDI, GDP, and Happiness. This reinforces the idea that health is not just a sector in isolation, but a central pillar of development. A population that lives longer and healthier tends to contribute more productively to the economy and experiences greater well-being overall. Recognising the importance of such features helps us go beyond the numbers and understand where investments and policy efforts could have the most profound impact. It reminds us that behind every data point is a story about how people live—and how they could live better.

Chapter 7:Impression Report

7.1 Personal Learning Experience

Working on this project was a way to apply what I've learned in both machine learning and development economics. I gained a deeper understanding of how real-world data can be used to model complex societal outcomes, and how machine learning can go beyond just predictions and into exploration and insight. It also helped strengthen my confidence in using tools like neural networks and evaluating their performance meaningfully.

7.2 Skill Enhancement

This project helped me sharpen several technical skills, especially in data preprocessing, feature selection, and building models for multi-output regression. I also became more comfortable using libraries like TensorFlow and Scikit-learn, and improved my ability to interpret model results. Beyond coding, I developed a better sense of how to communicate findings clearly and structure a complete data science workflow.

7.3 Industry/Academic Relevance

The work we did closely aligns with how machine learning is applied in both academic research and industry, especially in fields like policy modelling, international development, and social analytics. Predicting multiple development indicators and simulating "what-if" scenarios mirrors real-world problems tackled by economists, data scientists, and global organisations. The integration of explainability tools also reflects growing industry emphasis on interpretable AI.

Chapter 8: Conclusion & Future Work

This project was an attempt to bridge data science and real-world development analysis. Using both Random Forest and Neural Network models, we tried to predict key development indicators — like GDP per capita, HDI, MPI, and Happiness Score — based on various socio-economic inputs such as access to electricity, health and education spending, and GNI per capita.

While not everything went perfectly, the models performed quite well overall. The neural network, in particular, was able to pick up on complex, non-linear patterns that traditional methods might miss. That said, we also ran into some challenges, especially with data quality and consistency across countries. It became clear that no matter how advanced the model, it can only work as well as the data feeding into it.

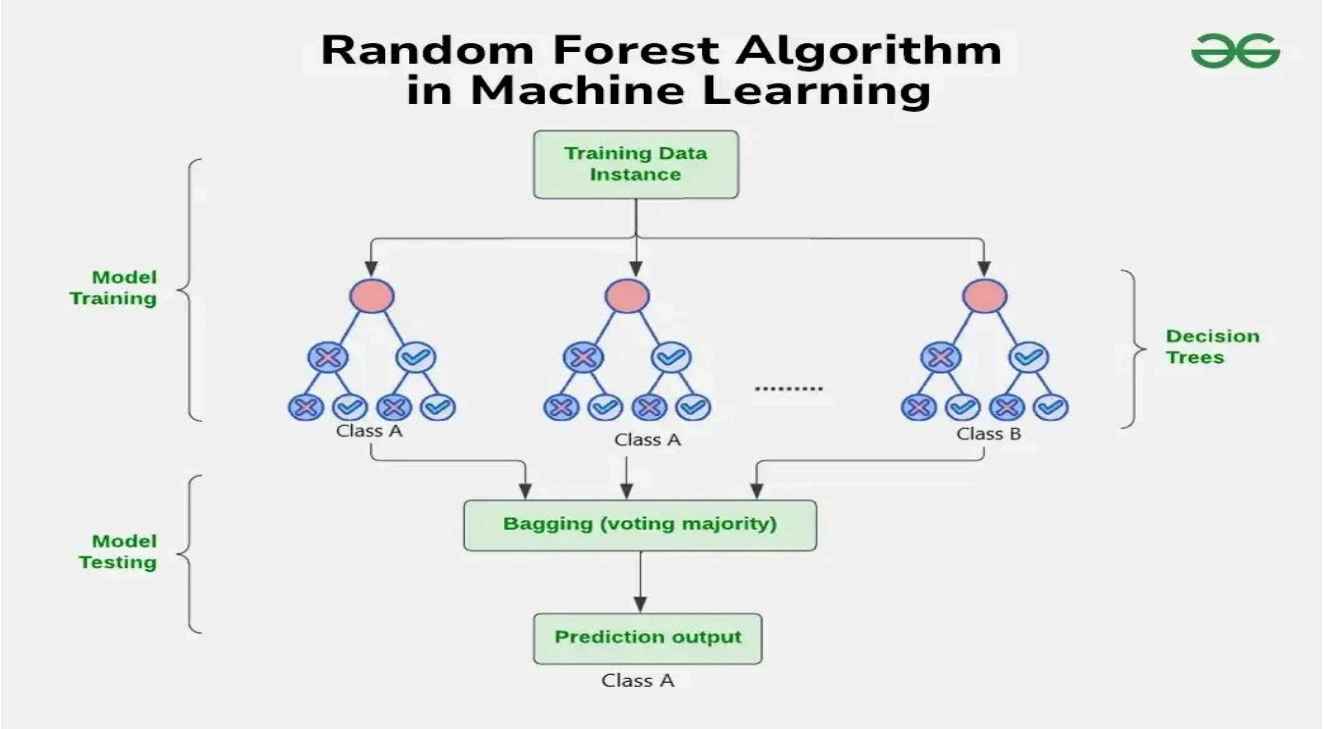
One of the more interesting takeaways was seeing which features had the most impact. Things like GNI per capita and social support came up repeatedly as strong predictors. These kinds of insights could be helpful for policymakers trying to figure out where to focus their efforts.

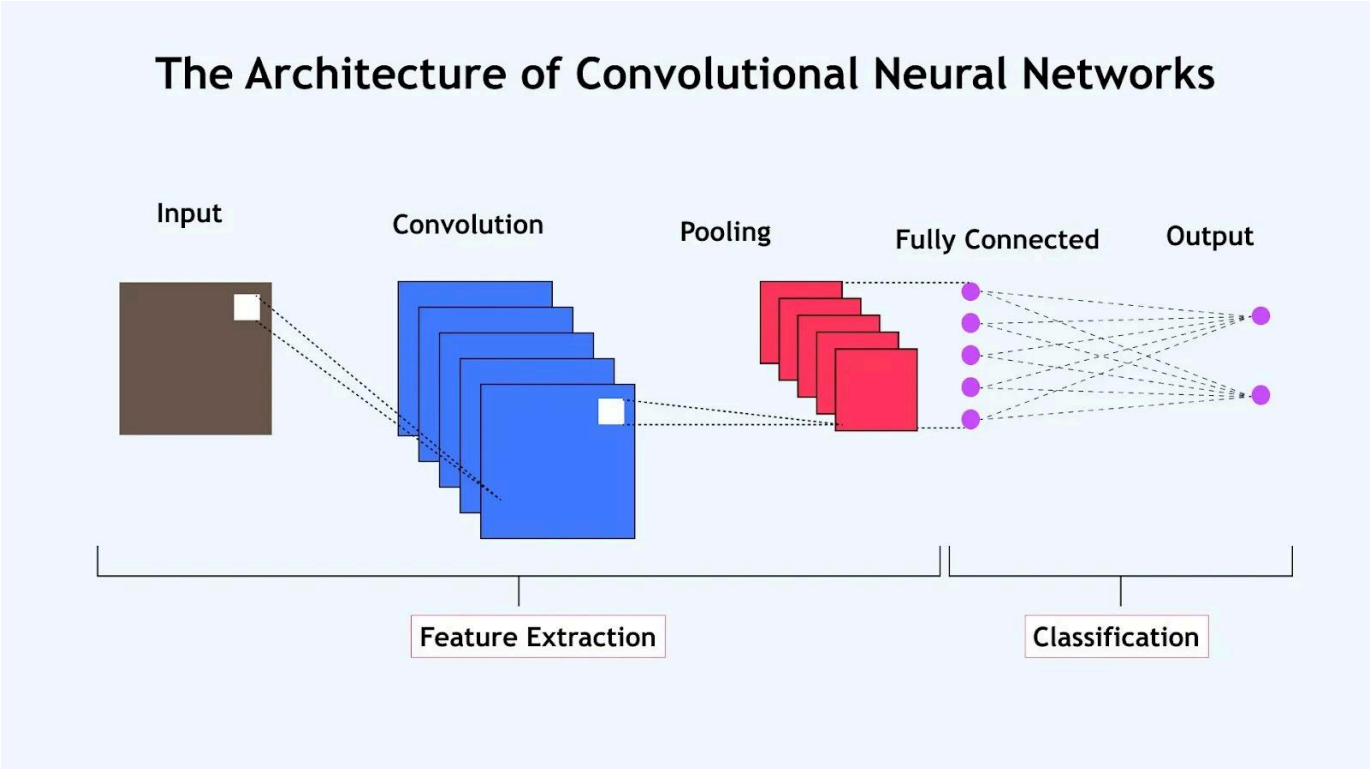
If we had more time, we would've loved to:

-Bring in more recent or detailed data (maybe even down to the regional level)
-Try out time-series models to predict changes over time
-Improve the neural network further by tuning its architecture
-Use explainability tools like SHAP more deeply to understand why the model makes certain predictions
-Build a simple interface that allows users to run their own "what-if" simulations

Overall, this project gave us a solid starting point. It showed that machine learning can absolutely be used to analyse development in a meaningful way, not just for prediction, but for exploration and insight too. There's still a lot more that can be done, but we've laid the groundwork for that next step.

Appendix






```
========= HDI =========
✅ Train R² = 0.9595 | Test R² = 0.8884 | Test MSE = 0.0022
📉 MAPE (HDI): 0.0490
```

```
========= GDP_PER_CAPITA =========
✅ Train R² = 0.9055 | Test R² = 0.7604 | Test MSE = 0.0139
📉 MAPE (gdp_per_capita): 338412842275.3156
```

```
🔍 Feature Importance:
Life expectancy at birth                    0.625250
Access to electricity (% of population)     0.277508
healthy_life_expectancy                     0.264011
generosity                                  0.093253
Unemployment rate (% of labor force)        0.081421
Mean years of schooling                     0.069299
Expected years of schooling                 0.067428
GNI per capita                              0.060368
freedom_to_make_life_choices                0.055099
social_support                              0.054576
Life expectancy                             0.041737
R&D spending (% of GDP)                     0.041582
Health spending (% of GDP)                  0.040717
Population growth (annual %)                0.037438
perceptions_of_corruption                   0.030466
Education spending (% of GDP)               0.017893
```

```
--- gdp_per_capita ---
R² Score: 0.8237144365622322
MSE: 0.037712630055311 79
MAPE: 17795737480898.863

--- HDI ---
R² Score: 0.7954204818208607
MSE: 0.005062020638495345
MAPE: 0.07446442388433015

--- happiness_score ---
R² Score: 0.8614703566389608
MSE: 0.17871671327730862
MAPE: 0.06307963622829323
```

REFERENCES

Mishra, P., & Nayak, S. (2024). *Machine Learning Projections for Human Development Index Anticipation*. ResearchGate. https://www.researchgate.net/publication/380589123_Machine_Learning_Projections_for_Human_Development_Index_Anticipation

Garcia, D., & Mavroeidis, S. (2020). *Predicting Economic Development Using Machine Learning*. Hugging Face. https://huggingface.co/papers/2010.10596

Sharma, M., & Bansal, S. (2022). Socio-Economic Indicator Forecasting using ML Techniques. In *2022 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)* (pp. 912–918). IEEE. https://ieeexplore.ieee.org/document/9733645

Wachter, S., Mittelstadt, B., & Russell, C. (2020). *Counterfactual Explanations and Algorithmic Recourse in Machine Learning*. arXiv. https://arxiv.org/abs/2010.10596

Xie, Y., Zhang, Y., Wang, Z., & Yu, D. (2024). Back to Basics: The Power of the Multilayer Perceptron in Financial Time Series Forecasting. *Mathematics, 12*(12), 1920. https://www.mdpi.com/2227-7390/12/12/1920